

MARKING CRITERIA

Important notes:

- The marking criteria is based on a detailed solution, but **some of the steps in the students' working might be implicit. If this is the case, marks need to be granted too.**
 - We **do not penalise a mistake twice**, i.e., if something was wrong in one part and was penalised, students can still get full marks in the other parts if they used their results consistently.
 - If students make a silly mistake when substituting the values (e.g., 60V rather than 600V), simply take out 0.5 marks and check the rest of the work accordingly. This is a substitution error rather than a flaw in the calculation method (e.g., overlooking a whole number in the calculation is not a silly mistake).
 - If students do **not** write **units in the final answer, take out 0.5 marks** in the corresponding question (only once per question).
 - Marks will not be granted for any of the formulae provided in the formula sheet, (i.e., writing a formula without any relevant context, pre-calculation, or substitution will not result in any mark).
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Question 1 (16 marks)

a. (9 marks)

i. (5 marks)

- **Note: Please note that it is not a requirement to draw the circuit.**
- Identifying that the “short-circuit” wire reconfigures the circuit into two parallel combinations -> 2 marks.
- Series/parallel simplification -> 1 mark each (3 in total)
Able to visualise $5//7//6$ and $10//5//3$ is in series with 2

ii. (2 marks)

- Calculation of the current -> 2 marks (1 mark for method, 1 mark for calculation and result)

iii. (2 marks)

- Calculation of the current -> 2 marks (1 mark for power calculation method, regardless of what method is used, and 1 mark calculation and result). Take out 0.5 marks if result is not in Joules (e.g., kWh)

b. (7 marks)

i. (4 marks)

- Identifying which LED is working (red) -> 1 mark
- Calculation of the current -> 2 marks (1 mark for equation, 1 mark for calculation and result)
- Academic advisor availability -> 1 mark (take out 0.5 marks if statement is not backed-up with calculations)

ii. (3 marks)

- Calculation of the current -> 1 mark (0.5 marks for equation, 0.5 marks for calculation and result – less marks because it is same process as in (i))
- Correct sign -> 1 mark
- Academic advisor availability -> 1 mark (take out 0.5 marks if statement is not backed-up with calculations)

Question 2 (22 marks)

Note: Students might label voltages/currents differently (i.e., + and – signs might not be the same as in the solution). Check the equations based on their labelling (explicit or implicit).

a. (12 marks)

i. (9 marks)

- KCL at node V1 -> 3 marks (1 mark per current in terms of nodal voltages – both value and sign need to be correct; if only KCL equation is provided correctly (e.g., $2 + i_2 - i_3 = 0$ for top node), 0.5 marks can be granted)
- KCL at node V2 -> 3 marks (1 mark per current in terms of nodal voltages; if only KCL equation is provided correctly, 0.5 marks can be granted)
- Calculate V1, V2 and V3 -> 3 marks (1 mark each)

ii. (1 mark)

- Calculate i_0 -> 1 mark (right or wrong)

iii. (2 marks)

- Power calculation -> 2 marks (1 mark for power calculation method, no matter what method is used, and 1 mark for calculation and result)

b. (10 marks)

- Identifying $I_3 = 3v_x \rightarrow 1$ mark
- Writing v_x in terms of mesh currents (might be implicit or explicit) and eliminating v_x from the mesh equations -> 1 mark
- Constraint equation -> 1 mark
- KVL in supermesh -> 6 marks (2 marks for identifying the supermesh, which might be implicit or explicit, 1 mark for each of the 3 resistor voltages in terms of the mesh currents, and 1 mark for the voltage across the voltage source)
- Simplification -> 1 mark

Question 3 (16 marks)

Note: If the student is using other methods rather than superposition, **ZERO** marks will be given.

a. (15 marks)

i. (15 marks)

- Contribution from the current source -> 6.5 marks
 - Using the right circuit (correctly turning off the voltage source, can be implicit or explicit) -> 1 mark
 - Calculation of i_{lights}' -> 5.5 marks (2 marks for sensible choice of method, 1.5 marks for equations (all of them need to be right to grant the mark), 1 mark for calculations and result).
- Contribution from the voltage source -> 6.5 marks
 - Using the right circuit (correctly turning off the current source) -> 1 mark
 - Calculation of i_{lights}'' -> 5.5 marks (2 marks for sensible choice of method, 1.5 marks for equations (all of them need to be right to grant the mark), 1 mark for calculations and result).
- Calculation of final result ($i_{lights} = i_{lights}' + i_{lights}''$) -> 2 marks

ii. (1 mark)

- Calculation of the power -> 1 mark (either right or wrong)

Question 4 (29 marks)

a. (12 marks)

- Source transformation (S.T.) -> 8 marks (4 S.T. in total with 2 marks assigned for each transformation: 0.5 marks for resistor value, 0.5 marks for source value, 1 mark for source polarity/direction)
- Simplification of sources -> 1 mark
- Simplification of resistors -> 1 mark (take 0.5 marks per mistake, up to a maximum of 1 mark)
- Calculation of R_{th} and V_{th} -> 2 marks

b. (17 marks)

i. (Thevenin voltage (8 marks))

Note: Students might label voltages/currents differently (i.e., + and – signs might not be the same as in the solution). Check the equations based on their labelling (explicit or implicit).

- Understanding open-circuit voltage calculation (i.e., open-circuit at terminals a-b – no current flowing through the 5-ohm, 8-ohm and 10-ohm resistor, $V_{th} = V_{oc}$) → 2 marks (might be implicit or explicit)
- Calculation of V_{th} -> 6 marks
 - Using Ohm's law
 - Source simplification (independent + dependent source) → 2 marks
 - Calculation of v_1 → 3 marks
 - 1.5 marks for independent source, 1.5 marks for dependent source
 - Calculation of V_{th} → 1 mark
 - Using KVL
 - KVL equation in the left mesh → 3 marks (might be implicit, 0.75 marks per voltage)
 - Calculation of v_1 → 2 marks
 - 1 mark for independent source, 1 mark for dependent source
 - Calculation of V_{th} → 1 mark
 - Using voltage division
 - Source simplification (independent + dependent source) → 2 marks
 - Voltage across 5-ohm resistor using voltage divider (in terms of v_1) → 2 marks
 - Expression of v_1 (might be implicit) to substitute in voltage divider formula → 1 mark
 - Calculation of V_{th} → 1 mark

ii. Thevenin resistor (7 marks)

Note: No mark is granted for stating $R_L = R_{th}$, since that formula is provided in the formula sheet.

Note: Method 1 is not valid in this question. Take out 4 marks if students turn off dependent source and use method 1 (i.e., 3 marks max. if they turn off all sources and calculate equivalent resistance).

Note: Even though the correct sign for R_{th} is negative, we will NOT penalise students if they work with the positive value.

- Using V_{th}/i_{sc} (method 2)
 - Short circuit the terminals a-b without turning off any source → 1.5 marks
 - Correct calculation of i_{sc} → 4 marks
 - Using mesh analysis
 - Mesh equations → 2 marks (deduct 0.5 marks per mistake up to a maximum of 2 marks)
 - Mesh currents calculation → 1.5 marks
 - Result (R_{th}) → 0.5 marks
 - $R_{th} = V_{th}/i_{sc}$ → 1.5 marks
- Using dummy current/voltage source (method 3)
 - Method for calculation → 3 marks (1.5 marks for adding dummy current or voltage source and using $R_{th} = V_0/1$ or $R_{th} = 1/i_0$, respectively, and 1.5 marks for turning off independent source)
 - Correct calculation of v_0 or i_0 → 4 marks
 - Using nodal analysis
 - Nodal equations → 2 marks (deduct 0.5 marks per mistake up to a maximum of 2 marks)
 - Nodal voltage calculation → 1.5 marks
 - Result (R_{th}) → 0.5 marks
 - Using mesh analysis
 - Mesh equations → 2 marks (deduct 0.5 marks per mistake up to a maximum of 2 marks)
 - Mesh currents calculation → 1.5 marks
 - Result (R_{th}) → 0.5 marks

iii. (2 marks)

Note: No mark is granted if the student only writes the formula without substituting the values, since the formula is given in the formula sheet.

Note: Even though the correct sign for R_{th} and power is negative, we will NOT penalise students if they work with the positive value.

- P_{max} calculation → 2 marks

Question 5 (17 marks)

i. Step response (7 marks)

- Identifying step response → 1 mark (might be explicitly stated or inferred from steps and/or formula used)
- Initial voltage $v(0) = 0$ → 1 mark
- Final voltage $v(\infty)$ → 2 marks (1 mark for open circuit voltage expression (which includes resistor combinations) and 1 mark for calculation)
- Time constant → 2 marks (1 mark for R_{th} and 1 mark for time constant)
- Step response → 1 mark for substitution, even if not fully explicit or simplified.

ii. Natural response (8 marks)

- Identifying natural response → 2 marks (might be explicitly stated or inferred from steps and/or formula used)
- Initial voltage $v(t_0)$ → 2 marks
- Time constant → 2 marks
- Natural response → 2 marks (1 mark for time shift and 1 mark for substitution, even if not fully explicit or simplified).

Note: If students flip step and natural response (sometimes because they wrongly understand how the switch works, sometimes because they do not fully understand the responses), they automatically lose 6 marks (for highlighted criteria). Then partial marks for each of the values (initial/final voltage and time constant) are granted as per the criteria above (i.e., if flipping the responses, students can get a maximum of 11 marks).

iii. Current at 5 minutes (2 marks)

- Current calculation (using $C \cdot dv/dt$ or $-v/12k$) → 2 marks